

Pure Substances and Mixtures — Paper B

MEDIUM • NON-CALCULATOR • 35 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided.
- No calculator is needed anywhere on this paper.
- Where a question says **explain**, your answer must say whether the atoms are **chemically joined** or **not joined**, and whether there is **one kind of particle** or more than one.
- The mark for each question is shown in square brackets on the right.
- Reminder: a pure substance melts **sharply** at one temperature; an impure one melts **over a range**, and lower.

SECTION 1 – COMPOUND AGAINST MIXTURE

- 1.** Complete the table comparing a compound with a mixture. Each empty box is worth one mark. [6]

	Compound	Mixture
Are the atoms chemically joined?		
Is the ratio of the substances fixed?		
Can it be separated by physical methods?		
Does it keep the properties of the substances in it?		no — new properties
Sharp melting point?		

- 2.** Sodium is a soft metal that fizzes violently in water. Chlorine is a poisonous green gas. Sodium chloride is the white solid you put on chips.

- (a)** State whether sodium chloride is an element, a compound or a mixture. [1]

- (b)** Explain why sodium chloride does not behave like sodium or like chlorine. [2]

SECTION 2 – IRON AND SULFUR

- 3.** A student mixes iron filings with sulfur powder, then heats half of the mixture in a test tube until it glows. The unheated half stays in a dish.

(a) A magnet is held over the dish. State what happens and explain why. [2]

(b) The magnet is held over the heated product. State what happens and explain why. [2]

(c) The student says the ratio of iron to sulfur in the heated product is "whatever we started with". Explain why this is wrong. [2]

SECTION 3 – TESTING FOR PURITY

4. Four samples labelled "benzoic acid" are tested. The data book gives the melting point of pure benzoic acid as **122 °C**.

Sample	Melting point
P	118 °C
Q	109–115 °C
R	122 °C
S	117–121 °C

(a) Which sample is pure benzoic acid? Give two reasons. [2]

(b) Which two samples are impure? How can you tell? [2]

(c) Sample P melts sharply but not at 122 °C. What does this tell you about P? [2]

5. Describe how you would test whether a sample of a solid is pure, and the result you would expect if it is. [3]

6. The nitrogen in air is not chemically joined to the oxygen. Give **two** pieces of evidence that air is a mixture. [3]

7. Explain why water is a pure substance even though it contains two different elements. [3]

End of paper. Check every answer has a unit or a form the question actually asked for.